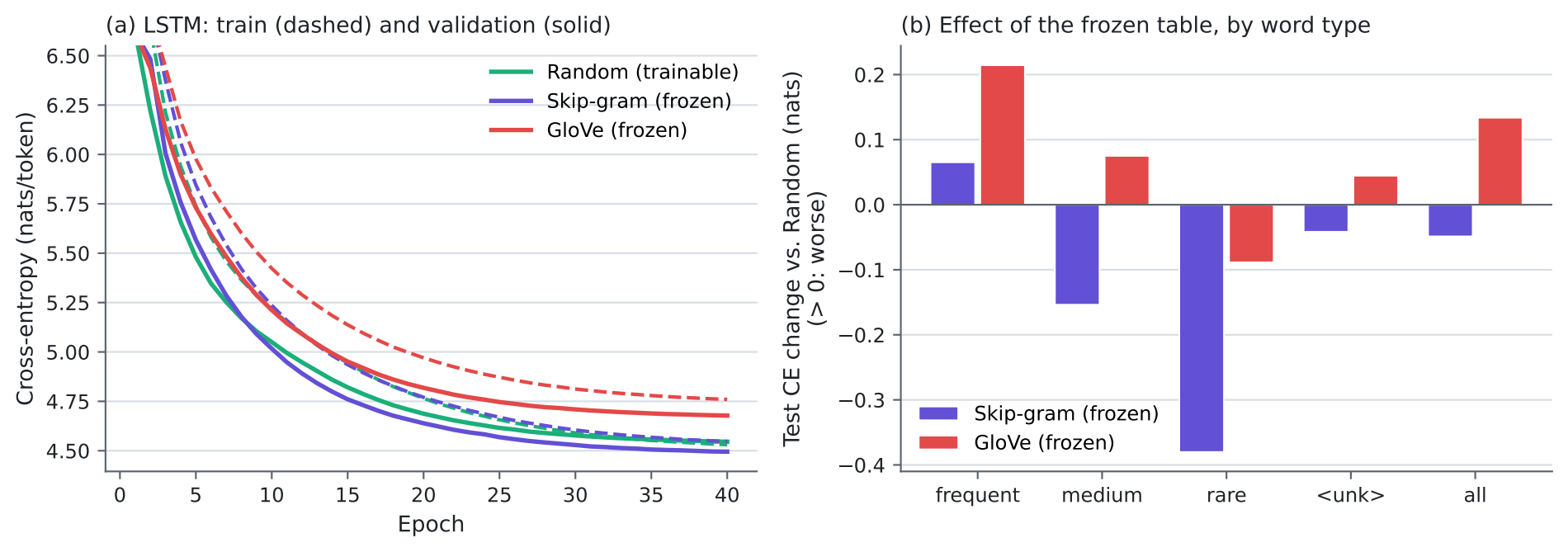
(a) LSTM: train (dashed) and validation (solid)
Cross-entropy (nats/token)
6.50
6.25
6.00
5.75
5.50
5.25
5.00
4.75
4.50
0
5
10
15
20
25
30
35
40
Epoch
Random (trainable)
Skip-gram (frozen)
GloVe (frozen)
(b) Effect of the frozen table, by word type
Test CE change vs. Random (nats)
(> 0: worse)
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
Skip-gram (frozen)
GloVe (frozen)
frequent
medium
rare
<unk>
all